

SPECIAL EVENT: PROFESSOR JENSEN'S POTIONS

DESCRIPTION: This event will be run at the ORLANDO tournament. This event is in memory of Mrs. Samantha Jensen, a supporter of Florida Science Olympiad and a coach from Orlando Science Schools. Ms. Jensen's passing was a huge loss to the Florida Science Olympiad community. Special themed awards will be given for this event. This event does not count toward overall team standings.

Teams will be asked questions and/or will conduct experiments at stations as they relate to **science laboratory equipment, chemical reactions, toxins and antidotes. Safety goggles are required.**

COMPETITORS BRING/NEED:

- Safety goggles
- Writing utensil;
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed, such as a computer, calculator or smartphone. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

NUMBER OF PARTICIPANTS: Maximum of 2

IMPOUND: NO

THE COMPETITION:

1. This event leans heavily on elements from the [Harry Potter](#) series, a favorite of Ms. Jensen.
2. Each team will move from one station to another. There will be 5 stations that can be completed in any order even though they have numbers. Each station should take approximately the same amount of time.

Station 1: First Year Potions Class

- a. This station focuses on activities related to lab safety and laboratory equipment such as:
 - i. What is the importance of PPE in the laboratory?
 - ii. What are standard safety expectations in a laboratory?
 - iii. What are the uses of common laboratory equipment, including graduated cylinders, beakers, flasks, hot water baths, bunsen burners, mortar and pestle, pipettes, pH paper, microscopes, test tubes, well plates, gloves, goggles, aprons, eye wash stations, etc..

Station 2: Second Year Potions Class

- a. At Hogwarts, second-year students focused on a specific element of potions and brewed simple potions.
- b. This station focuses on simple chemical reactions that include Sodium Bicarbonate (baking soda), Acetic Acid (vinegar), Calcium Carbonate (egg shells), Calcium Chloride, water, and/or Ascorbic Acid (Vitamin C). Sample tasks may include:
 - i. Describing the physical and chemical properties of the compounds listed above.
 - ii. Performing simple experiments and collecting data using the compounds above
 - iii. Identifying the characteristics of a chemical change
 - iv. Writing simple chemical equations using the compounds listed above.

Station 3: Third Year Potions Class

- a. At Hogwarts, third-year students focused on "Undetectable Poisons".
- b. This station focuses on toxic plants. Students will need to be able to identify these plants from images, explain the symptoms of poisoning from these plants, and describe possible remedies/antidotes. The plants that students may be tested on are:
 - i. Wolfsbane (*aconitum sp.*)
 - ii. Jack-in-the-pulpit (*Arum maculatum*)

- iii. Lily of the Valley (*Convallaria majalis*)
- iv. Poison sumac (*Toxicodendron vernix*)
- v. Poison Ivy (*Toxicodendron radicans*)
- vi. Water Hemlock (*Cicuta maculata*)
- vii. Tobacco (*Nicotiana tabacum*)
- viii. Mistletoe (*Viscum album*)

Station 4: Fourth Year Potions Class

- a. At Hogwarts, fourth-year students focused on “Poison Antidotes”
- b. This station focuses on toxic animals. Students will need to be able to identify these animals from images, explain the symptoms of poisoning/envenomation from these animals, and describe possible remedies/antidotes. The animals that students may be tested on are:
 - i. Poison dart frogs (*Dendrobates sp*)
 - ii. Portuguese man o’war (*Physalia physalis*)
 - iii. Fattail scorpion (*Androctonus australis*)
 - iv. Cone snails (*Class Conoidea*)
 - v. Cane toad (*Rhinella marina*)
 - vi. Box jellyfish (*Carukia barnesi*)
 - vii. Fire salamander (*Salamandra salamandra*)
 - viii. Olive Sea Snake (*Aipysurus laevis*)

Station 5: Fifth Year Potions Class

- a. At Hogwarts, fifth-year students focused on properly brewing potions using directions.
- b. Students will be asked to complete a lab activity following directions from the supervisor which may include:
 - i. Mixing reagents and measuring: reaction rate, production of gas, color change, development of a precipitate
 - ii. Separating mixtures
 - iii. Determination of pH

RELATED FLORIDA STANDARDS:

- **SC.5.P.8.1** Compare and contrast the basic properties of solids, liquids, and gases, such as mass, volume, color, texture, and temperature
- **SC.3.P.8.1** Measure and compare temperatures of various samples of solids and liquids.
- **SC.3.P.8.2** Measure and compare the mass and volume of solids and liquids.
- **SC.5.P.8.3** Demonstrate and explain that mixtures of solids can be separated based on observable properties of their parts such as particle size, shape, color, and magnetic attraction.
- **SC.5.P.8.2** Investigate and identify materials that will dissolve in water and those that will not and identify the conditions that will speed up or slow down the dissolving process.
- **SC.5.P.9.1** Investigate and describe that many physical and chemical changes are affected by temperature.

SCORING:

1. **High score wins.**
2. Each question or station will be assigned a predetermined set of points, assigned by the event supervisor **and communicated to the teams (preferably- points are identified on the test/station paperwork).**
3. Ties will be broken by predetermined questions.